A c0 Counterexample to Aggregate Domination for Independent Sums

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Status

This packet gives a candidate counterexample, likely valid, to the discrete problem stated in Remark 12.10 of Yaroslavtsev's paper *Local characteristics and tangency of vector-valued martingales* [1]. The result refutes the broad Banach-space problem as stated there. It does not settle the corresponding UMD-space problem for general characteristically dominated local martingales, since the counterexample uses $X = c_0$, which is not UMD.

Source Problem

The source is arXiv:1907.11588, PDF page 105, Remark 12.10. The source crop below shows the formal statement.

(12.11), (12.14), (2.9), and the fact that M and N are quasi-left continuous. $\square$

Remark 12.10. It is not known whether Theorem 12.8 holds for general local martingales. By Theorem 3.39 and by Proposition B.1 the main issue here is in proving a similar statement for discrete martingales with independent increments. Let us state this problem here as open. Let X be a Banach space, $1 \leq p < \infty$. Let $(\xi_n)_{n\geq 1}$ and $(\eta_n)_{n\geq 1}$ be X-valued mean-zero independent random variables such that for any Borel $B \in X \setminus \{0\}$

$$\sum_n \mathbb{P}(\xi_n \in B) \leq \sum_n \mathbb{P}(\eta_n \in B).$$

Does there exists a constant C (perhaps depending on p and X) such that

$$\mathbb{E}\Big\|\sum_n \xi_n\Big\|^p \leq C\mathbb{E}\Big\|\sum_n \eta_n\Big\|^p?$$

The next page adds that, by symmetrization, one can assume the variables are symmetric, and that even this case was not known to the author.

> By a standard symmetrization trick [66, Lemma 6.3] one can assume that ξ_n's and η_n's are symmetric. But even the symmetric case is not known for the author.

In modern notation, the question asks: if X is a Banach space, $1 \le p < \infty$, and (ξ_n), (η_n) are independent mean-zero X-valued random variables satisfying

$$\sum_n \mathbb{P}(\xi_n \in B) \le \sum_n \mathbb{P}(\eta_n \in B) \qquad \text{for every Borel } B \subset X \setminus \{0\},$$

must there be a constant $C = C(p, X)$ such that

$$\mathbb{E}\Big\|\sum_n \xi_n\Big\|^p \le C\,\mathbb{E}\Big\|\sum_n \eta_n\Big\|^p?$$

Idea of the Proof

Aggregate domination records only the total amount of mass assigned to each jump vector. It forgets how that mass is grouped into independent variables. In c_0, this grouping matters. If the first N signed basis vectors are used once each, the sum has sup norm exactly 1. If instead one takes N independent uniformly chosen signed basis vectors among the first N coordinates, then the aggregate jump measure is identical, but the sup norm of the sum is the largest signed occupancy count among N boxes. That maximum diverges.

Counterexample

Theorem 1. *For every $1 \le p < \infty$, there is no constant $C = C(p, c_0)$ satisfying the inequality in Remark 12.10 for all finite independent symmetric c_0-valued families. This remains true even when the aggregate measures $\sum_n \mathbb{P}(\xi_n \in \cdot)$ and $\sum_n \mathbb{P}(\eta_n \in \cdot)$ are equal on $c_0 \setminus \{0\}$.*

Proof. Let $(e_k)_{k\ge1}$ be the canonical unit vector basis of c_0. Fix $N \ge 2$. Let $\varepsilon_1, \dots, \varepsilon_N$ be independent Rademacher signs and set

$$\eta_k = \varepsilon_k e_k, \qquad 1 \le k \le N.$$

Let $I_1, \dots, I_N$ be independent uniform random variables on $\{1, \dots, N\}$, independent of another Rademacher family $\delta_1, \dots, \delta_N$, and set

$$\xi_j = \delta_j e_{I_j}, \qquad 1 \le j \le N.$$

All these variables are independent and symmetric, hence mean zero.

For every Borel set $B \subset c_0 \setminus \{0\}$,

$$\sum_{j=1}^N \mathbb{P}(\xi_j \in B) = N \cdot \frac{1}{2N}\big(\#\{1 \le k \le N:\ e_k \in B\} + \#\{1 \le k \le N:\ -e_k \in B\}\big) = \sum_{k=1}^N \mathbb{P}(\eta_k \in B).$$

Thus the hypothesis holds with equality.

On the other hand,

$$\Big\|\sum_{k=1}^N \eta_k\Big\|_\infty = 1 \quad \text{almost surely.}$$

It remains to show that

$$\mathbb{E}\Big\|\sum_{j=1}^{N}\xi_j\Big\|_\infty^p \to \infty.$$

Let

$$S_N = \sum_{j=1}^{N}\xi_j, \qquad r_N = \left\lfloor \frac{\log N}{4\log\log N}\right\rfloor$$

for N sufficiently large. For each $1 \le k \le N$, let A_k be the event that exactly r_N of the variables ξ_j are equal to $+e_k$ and none is equal to $-e_k$. If A_k occurs, then the k-th coordinate of S_N is r_N, so $\|S_N\|_\infty \ge r_N$.

Put $r = r_N$ and $Z_N = \sum_{k=1}^N \mathbf{1}_{A_k}$. For a fixed k,

$$p_N := \mathbb{P}(A_k) = \binom{N}{r}\left(\frac{1}{2N}\right)^r\left(1-\frac{1}{N}\right)^{N-r}.$$

Since $r = o(\sqrt{N})$,

$$p_N \ge \frac{c}{2^r r!}$$

with an absolute constant $c > 0$, for all large N. Hence

$$\mathbb{E}Z_N = Np_N \ge c\,\frac{N}{2^r r!} \to \infty,$$

because

$$\log(2^r r!) = r\log r + O(r) = \left(\frac{1}{4} + o(1)\right)\log N.$$

For $k \ne \ell$,

$$q_N := \mathbb{P}(A_k \cap A_\ell) = \frac{N!}{r!\,r!\,(N-2r)!}\left(\frac{1}{2N}\right)^{2r}\left(1-\frac{2}{N}\right)^{N-2r}.$$

A direct comparison with p_N^2 gives

$$\frac{q_N}{p_N^2} = \frac{(N-r)!^2}{N!(N-2r)!}\,\frac{(1-2/N)^{N-2r}}{(1-1/N)^{2N-2r}} \le C_0$$

for all large N, since $r = o(\sqrt{N})$. Therefore

$$\mathbb{E}Z_N^2 = Np_N + N(N-1)q_N \le \mathbb{E}Z_N + C_0(\mathbb{E}Z_N)^2 \le C_1(\mathbb{E}Z_N)^2$$

for all large N. By the Paley–Zygmund inequality,

$$\mathbb{P}(Z_N > 0) \ge \frac{(\mathbb{E}Z_N)^2}{\mathbb{E}Z_N^2} \ge C_1^{-1}.$$

Consequently,

$$\mathbb{E}\|S_N\|_\infty^p \ge r_N^p\,\mathbb{P}(Z_N > 0) \ge C_1^{-1}\left\lfloor\frac{\log N}{4\log\log N}\right\rfloor^p \to \infty.$$

Thus the ratio

$$\frac{\mathbb{E}\|\sum_{j=1}^N \xi_j\|_\infty^p}{\mathbb{E}\|\sum_{k=1}^N \eta_k\|_\infty^p}$$

is unbounded. No constant $C(p, c_0)$ can exist. $\square$

Verification Notes

The proof uses only finite sequences, so there is no convergence issue in the sums. The variables are symmetric, so the counterexample also refutes the symmetric version mentioned immediately after Remark 12.10. The same construction lives inside c_0 for all N, so the failure is for one fixed Banach space, not for a changing finite-dimensional space.

The small script `code/occupancy_lower_bound.py` prints the first- and second-moment quantities used in the Paley–Zygmund estimate for sample values of N. It is only a sanity check; the preceding occupancy estimate is the proof.

Novelty and Literature Check

The run indexes were searched for `1907.11588`, `characteristically dominated`, `recoupling`, `UMD+`, and the discrete aggregate-domination formulation; no prior run packet for this target was found. Web searches were run for exact and nearby phrases including "It is not known whether characteristically dominated martingales", "characteristically dominated martingales independent random variables", and "sum_n P(xi_n in B) sum_n P(eta_n in B) Banach". These found the source paper and related comparison literature, including Montgomery-Smith–Pruss [2], but no existing answer to this aggregate-domination formulation.

Limitations

This is a counterexample to the broad Banach-space discrete question in Remark 12.10. It does not resolve whether the desired comparison might hold under additional geometric assumptions, especially for UMD spaces. Therefore it does not by itself settle the full general-local-martingale characteristically dominated problem in the UMD setting.

Human Review Recommendation

Review the second-moment occupancy estimate and the exact matching of aggregate measures. If those two points check out, the counterexample should be accepted as a negative answer to the Banach-space discrete question exactly as stated in Remark 12.10.

References

[1] I. S. Yaroslavtsev, *Local characteristics and tangency of vector-valued martingales*, arXiv:1907.11588, 2019.

[2] S. J. Montgomery-Smith and A. R. Pruss, *A comparison inequality for sums of independent random variables*, J. Math. Anal. Appl. 254 (2001), no. 1, 35–42; arXiv:math/9811124.